

Neural Networks vs. Classical Calibration of the SABR Model

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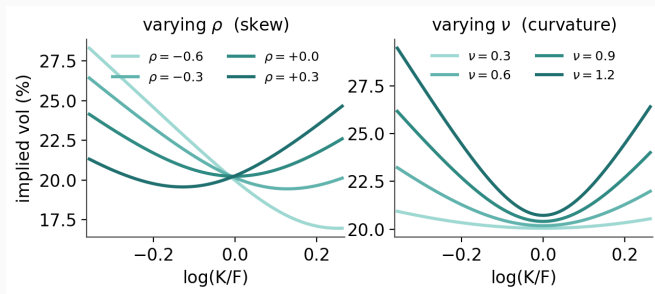
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Motivation

The SABR model

SABR (Hagan et al., 2002)
describes the volatility *smile* with
four parameters:

- α — initial volatility level
- β — backbone elasticity
- ρ — correlation (the skew)
- ν — vol-of-vol (the curvature)



ρ tilts the skew, ν sets the curvature.

How calibration is done

Calibration recovers (α, ρ, ν) from a single observed smile. The elasticity β is not fitted: it is fixed to $\beta = 1$ in advance, because from one smile β and ρ produce near-identical skews — fitting both is ill-posed, so β is pinned and only ρ is left free.

Two conventions that shape the comparison

- **Per-smile, not whole-surface** — one (α, ρ, ν) triple per (date $\times$ expiry), with no information shared across maturities.
- **Repeated constantly** — every underlying $\times$ expiry $\times$ trading day is a fresh fit.

The standard recipe: evaluate Hagan's closed-form expansion for σ inside a bounded least-squares loop (`scipy.trf`).

Prior work, and what is new here

Prior work

Horvath (2021), Stuijt (2021) replace the Hagan formula with a trained **NN surrogate** of the vol map — motivated by *evaluation speed*.

But the analytic formula is *already* cheap, so speed is a weak motivation. Its real weakness is **non-uniform error**: Hagan drifts in the *wings* (Antonov et al., 2013) and at *short maturities* (Obloj, 2008) — exactly where a high vol-of-vol regime bites.

What is new

1. **Sobolev loss** — match Hagan's values *and* its parameter-gradients
2. **Activation selected**, not assumed — four-way over Tanh / ELU / GELU / SIREN
3. **Conditioning and Greeks measured**, not only the volatility fit

This turns a one-axis question into three:

fit
σ-RMSE

conditioning & Greeks
 $\kappa(J), \partial\sigma$

cost
warm-start

Three research questions

1. Can a surrogate *trained on Hagan* match or beat Hagan when both are scored against *real market quotes*?
2. Does smoothness regularisation deliver better-conditioned Jacobians, and do its Greeks agree with the closed form?
3. Can a warm-start hybrid cut the calibration's iteration count without surrendering precision?

Methodology

Three models

1. **Classical baseline** — Hagan inside bounded LSQ (`trf`), analytic torch-autograd Jacobian. The reference for every comparison.
2. **Surrogate NN** — a 5×32 MLP approximating the Hagan vol map, **dropped in** as the `vol_fn` in place of the analytic formula.
3. **Warm-start NN** — a second net predicts the **starting parameters** (α, ρ, ν) for the optimiser, and Hagan stays inside the loop.

The surrogate, and how it is trained

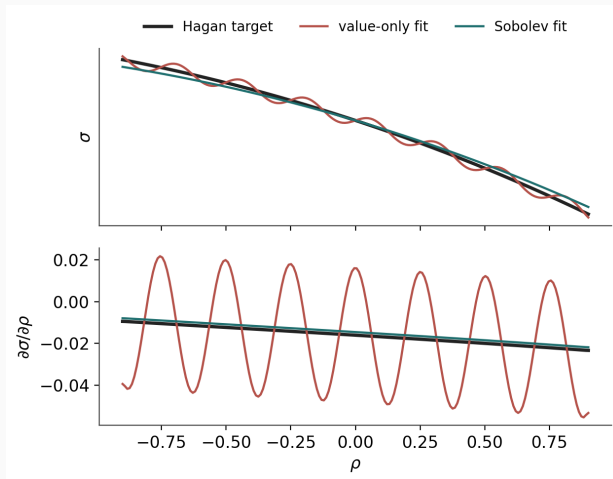
Architecture

- 5×32 MLP, **4,449** params
- $(m = \log \frac{F}{K}, T, \alpha, \rho, \nu) \rightarrow \sigma$
- pointwise — the smile is assembled strike by strike

Sobolev loss

$$\mathcal{L} = \underbrace{\|\sigma_\theta - \sigma_H\|^2}_{\text{value}} + \lambda \underbrace{\|\nabla_\theta \sigma_\theta - \nabla_\theta \sigma_H\|^2}_{\text{slope } (\partial/\partial\theta)}$$

Matches the value *and* the slope —
smooth in the sense the calibrator uses.

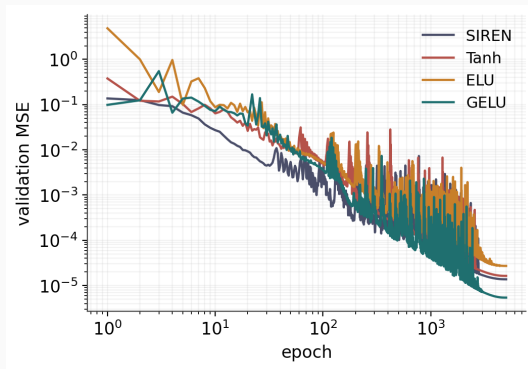


Comparing the four activations

Tanh, ELU, GELU, SIREN — each trained to convergence (5000 ep, cosine LR + AdamW, best-val checkpoint).

Activation	synthetic val-MSE
GELU	8.4×10^{-6}
SIREN	2.0×10^{-5}
Tanh	2.1×10^{-5}
ELU	2.7×10^{-5}

Caveat: this synthetic ranking does *not* predict the real-data ranking — model selection must use real-data fit.



Convergence trajectories per activation.

Data

Gold options, two windows

Underlying: SPDR Gold Shares (GLD).

Primary — Feb–Mar 2026 spike

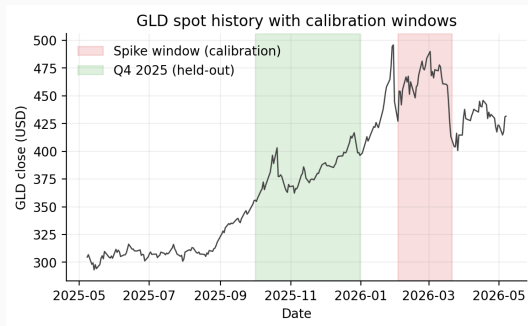
high vol-of-vol, asymmetric wings.

234 smiles (one per date $\times$ expiry).

Holdout — Q4 2025

calmer pre-spike regime, no retraining. 137 smiles for cross-window robustness.

Source: Polygon.io free tier (OHLCV). BS IV inverted locally against a parity-derived forward. (No NBBO mid — an honest data caveat.)



GLD spot — the two calibration windows.

Results

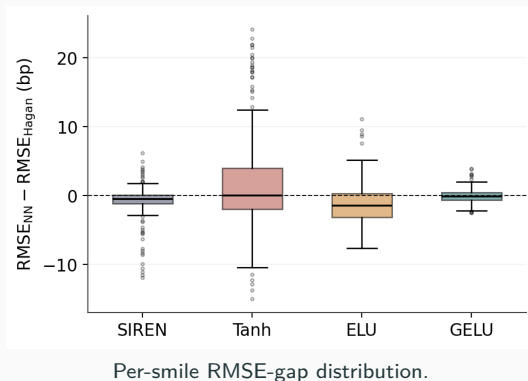
Axis 1 — fit against the market

Median per-smile RMSE gap to Hagan, spike window (negative = surrogate better):

Act.	median gap	Wilcoxon p
ELU	-1.47 bp	1.4×10^{-11}
SIREN	-0.49 bp	1.4×10^{-12}
GELU	-0.13 bp	2.8×10^{-2}
Tanh	+0.03 bp	n.s.

3 of 4 at or below Hagan at the median. The gain sits in the *wings*, where Hagan's expansion is weakest.

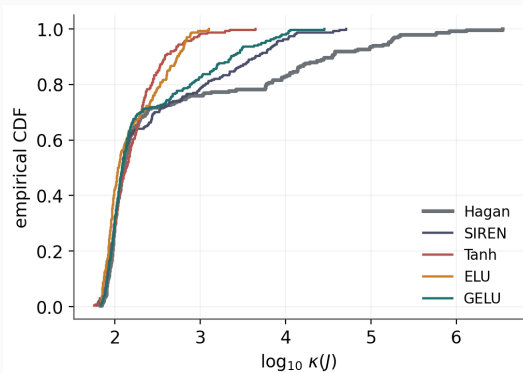
On a minority of sharply-curved smiles the surrogate fits worse than Hagan (worst gap +3.9 bp GELU, +24 bp Tanh) — the tail that lifts Tanh's mean.



Axis 2 — Jacobian conditioning $\kappa(J)$

$\kappa(J)$ is the **condition number** of the calibration Jacobian — how much the fitted (α, ρ, ν) amplify small errors in the quotes. **Lower is better**; a large $\kappa(J)$ is a near-singular fit that noise can swing wildly.

- **Median:** ELU's median $\kappa(J)$ is $0.86\times$ Hagan's, and the other three land within 7% of Hagan ($0.93\text{--}1.07\times$).
- **Distribution** (stronger): **every** activation's $\kappa(J)$ stays bounded by Hagan in *both* windows.



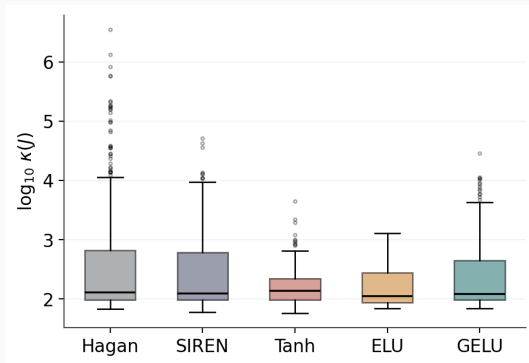
Cumulative distribution ($\log_{10} \kappa(J)$, 234 spike smiles): the share of smiles below each value. Every NN curve saturates left of Hagan's $\sim 10^{6.5}$ tail.

Axis 2 — and the heavy tail is suppressed

The headline is in the **worst case**, not the median:

- Hagan's $\max \kappa(J) = 3.5 \times 10^6$
- every NN max-tail is $\geq$ **one order of magnitude below** (ELU tightest, 1.3×10^3)
- the surrogate **removes Hagan's heavy upper tail** of ill-conditioned smiles

The smooth surrogate never produces the near-singular fits that Hagan's algebraic expansion occasionally does.



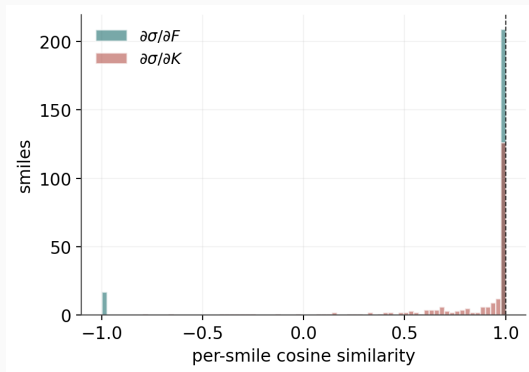
Per-smile $\kappa(J)$ ($\log_{10}$): Hagan's box and whisker run far higher than any activation's.

Axis 2 — and the Greeks agree

Each method autograded at *its own* calibrated (α, ρ, ν) . Per-strike $\partial\sigma/\partial F$, $\partial\sigma/\partial K$:

- median cosine **1.0000** / **0.9882** (spike),
1.0000 / 0.9996 (Q4)
- relative MAE 12% / 17% of Greek scale
(spike), falling to 1–3% on Q4

The small cluster at cosine -1 is a **near-zero artefact**: where $\partial\sigma/\partial F$ is itself ≈ 0 , its sign carries no information — not a real disagreement.



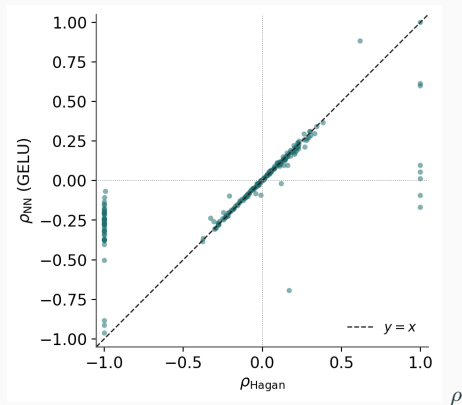
Per-smile Greek-agreement distribution (spike).

Axis 2 — where they disagree: ρ at the boundary

The only real disagreement sits on the smiles where Hagan is **driven to** $\rho = \pm 1$ while the NN settles *interior* (the vertical bands).

There the two methods fit the market σ comparably but recover *different* ρ , so their $\partial\sigma/\partial K$ profiles diverge.

This single effect is the entire $\partial\sigma/\partial K$ disagreement tail above.



recovered by the NN vs Hagan, 234 spike smiles.

Axis 3 — warm-start speed

A net predicts the starting parameters, and Hagan stays the inner vol-function.
Significance by **Wilcoxon signed-rank** on paired per-smile nfev.

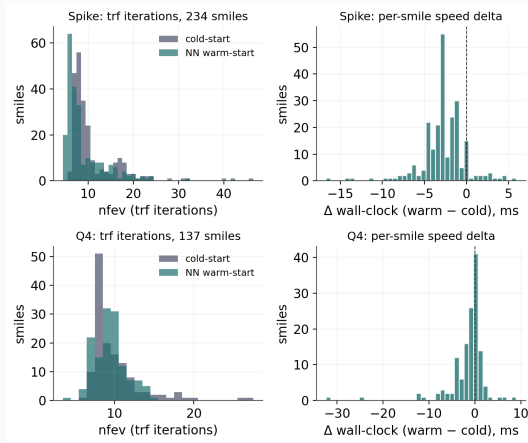
Spike (234 smiles)

- median nfev $9 \rightarrow 7$ (**22%**),
 $p = 8.1 \times 10^{-21}$
- $1.25\times$ wall-clock speedup, **no precision loss**

Q4 holdout (137 smiles)

- median nfev $9 \rightarrow 9$ — **no median reduction**
- mean reduction only 5.6%, $p = 0.21$ (**not significant**)

The deployable speed claim is therefore *spike-only*.



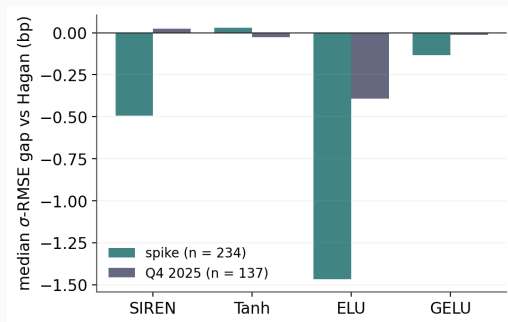
nfev distribution: cold vs warm start.

Does it survive the calmer regime? (Q4 holdout)

No retraining — the spike result re-tested on Q4:

- σ -fit: median gaps within ± 0.4 bp of zero, tails contract.
- $\kappa(J)$: **bounded-tail behaviour holds** — every NN still dominated by Hagan's tail.
- warm-start: advantage **attenuates** (regime-dependent margin).

The **fit** and **conditioning** results carry over to the calmer regime. The warm-start speed-up does *not* — it is the one spike-specific finding.



Spike vs Q4 side-by-side.

Conclusion

What the comparison establishes

Within a deliberately narrow scope — basic Hagan baseline, $\beta = 1$ per-smile fit, two windows, one underlying, free-tier data — the surrogate gives measurable improvements on **three axes**:

Fit

3 / 4 activations at or below Hagan's median σ -RMSE on real quotes.

Conditioning

$\kappa(J)$ stays below Hagan, and the rare near-singular fits Hagan produces disappear.

Speed

22% fewer iterations via warm-start (spike), no precision loss.

Limitations, and the larger question

Honest scope

- Baseline is *basic* Hagan — no wings-corrected or short-time baseline. A corrected Hagan would likely narrow the σ -fit gap.
- Two windows, one underlying. The smiles are not iid, so p -values overstate the effective sample size.
- Warm-start speed is spike-only.

The research question this motivates

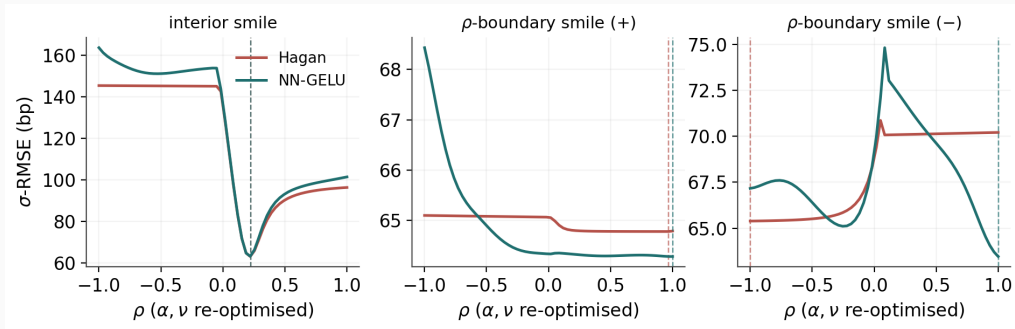
Each axis covers ground a theoretician's *analytical* fix has historically occupied. **Can NN tooling systematically reproduce SABR's hand-derived corrections** — wings, short-time, negative-rate, multi-smile — from training alone? What this thesis shows on one model and one underlying, the broader programme would test against each correction the SABR literature has derived by hand.

Thank you.

Questions welcome.

Backup — the ρ -recovery diagnostic

On boundary-affected smiles the NN recovers an *interior* ρ where Hagan is driven to $\rho = \pm 1$. Is the smooth class *unable* to reach the boundary, or does the calibrator just not get there? Profiling the loss in ρ (α, ν re-optimised at each ρ) settles it.



On the ρ -boundary smiles the surrogate's own loss minimum *can* sit at the boundary (left of each dashed line). The gap is **convergence, not representation** — the fix is a better starting point, not a bigger network.

Backup — the SIREN narrow-sampler failure

An early SIREN-only net (narrow bounds $\nu \leq 1.5$, constant LR, 500 ep) gave a +6.45 bp median gap and pinned $\rho \approx 0$.

Diagnosis: 34% of real calibration probes asked for $\nu > 1.5$ — extrapolation off the training support. The θ -Jacobian degraded there and `trf` stalled at the $\rho \approx 0$ saddle.

Fix (concurrent): widen the sampler, cosine LR + λ -warmup + best-val checkpoint, all four activations to 5000 ep.

Lesson: extrapolation fails at the *calibrator* level, not the network output — predicted σ looks fine past the boundary, but its parameter-gradients have no training signal.

Backup — the Wilcoxon signed-rank test

Every per-smile comparison is **paired**: the same smile is fit by Hagan and by the surrogate, so the natural statistic is the per-smile *difference* (e.g. σ -RMSE gap, or cold – warm nfev).

- The **Wilcoxon signed-rank** test asks whether those paired differences are symmetric about zero — a non-parametric paired test, no normality assumption.
- Chosen over the paired t -test because the gap distributions are heavy-tailed (Table of statistics) and the median, not the mean, is the headline.
- Caveat: the smiles share trading days and chains, so they are not iid — the reported p -values *overstate* the effective sample size.

Backup — what $\kappa(J)$ is, and why it matters

At the optimum, `trf` works with the Jacobian $J = \partial r / \partial (\alpha, \rho, \nu)$ of the residuals $r = \sigma_{\text{model}} - \sigma_{\text{market}}$. Its condition number

$$\kappa(J) = \sigma_{\max}(J) / \sigma_{\min}(J)$$

is the ratio of largest to smallest singular value.

- $\kappa(J)$ measures how strongly a small error in the quotes is **amplified** into the recovered parameters — the local sensitivity of the fit.
- A large $\kappa(J)$ means a near-flat direction in parameter space (e.g. the β - ρ or boundary- ρ degeneracies): the fit is unstable and noise-sensitive.
- **Lower is better.** The surrogate's smoothness bounds $\kappa(J)$ below Hagan's and removes its heavy ill-conditioned tail.

Backup — distribution statistics (spike, per smile)

σ -RMSE gap to Hagan (bp) <i>negative = surrogate better</i>							
Act.	median	mean	std	min	q05	q95	max
SIREN	−0.49	−0.83	2.34	−11.92	−5.33	+1.85	+6.23
Tanh	+0.03	+1.70	7.21	−14.99	−8.66	+18.35	+24.15
ELU	−1.47	−1.24	3.00	−7.66	−5.53	+3.80	+11.13
GELU	−0.13	−0.08	1.10	−2.51	−1.96	+1.81	+3.94

$\kappa(J)$ ratio, NN / Hagan $< 1 = \text{surrogate better-conditioned}$							
Act.	median	mean	std	min	q05	q95	max
SIREN	0.914	0.908	0.652	2.6e−4	0.036	1.63	7.25
Tanh	0.951	0.773	0.449	9.5e−5	0.005	1.32	1.79
ELU	0.808	0.740	0.542	7.3e−5	0.003	1.17	5.10
GELU	0.941	0.785	0.389	2.5e−4	0.029	1.05	3.03